QA Embedded — Hardware Test Sheet / Test Log

Smart Gate Controller • Wokwi • Black-box testing

Test Session

Tester		Date	
Alexandra Holoborodko		24-26.09.2026	
Build / Firmware	Build_1	Environment	Wokwi
Project	Smart Gate Controller	Board	Arduino Uno
Test Type	Functional / Regression / Exploratory	Result	Rejected for release

Test Execution Rules

- Test the device as a black box.
- Record observable behavior, not assumptions about the implementation.
- For every FAIL, create a Bug Report.
- Use the same test cases for regression after firmware changes.
- Try both positive and negative scenarios.

Test Cases

ID	Requirement	Precondition	Action / Steps	Expected Result	Actual Result	P/F	Bug ID/comment
TC-01	R1	After startup	Start simulation	State CLOSED; relays OFF	State CLOSED; relays OFF	Passed	
TC-02	R2	CLOSED	OPEN	OPENING; OPEN relay ON; OPEN LED ON	OPENING; OPEN relay ON; OPEN LED ON	Passed	
TC-03	R3	OPENING	Activate LIMIT_OPEN	OPEN; OPEN relay OFF	OPEN LIMIT DETECTED in Terminal, OPEN LED ON OPEN relay = OFF; CLOSE relay = OFF;	Failed	Bug_1 R6 req? R15 req?
TC-04	R4	OPEN	CLOSE	CLOSING; CLOSE relay ON; CLOSE LED ON		BLOCKED (by TC_3)	R3 req?
TC-05	R5	CLOSING	Activate LIMIT_CLOSE	CLOSED; CLOSE relay OFF	CLOSED; CLOSE relay OFF OPEN relay OFF	Passed	R7 req? R15 req?
TC-06	R6	OPENING	STOP	STOPPED; both relays OFF	STOPPED; both relays OFF	Passed	R5 req?
TC-07	R6	CLOSING	STOP	STOPPED; both relays OFF	STOPPED; both relays OFF	Passed	R5 req?
TC-08	R7	OPENING	OPEN again	No unsafe/contradictory state	IGNORED: OPEN is not valid in current state	Passed	R13 req?
TC-09	R7	CLOSING	CLOSE again	No unsafe/contradictory state	IGNORED: CLOSE is not valid in current state	Passed	R13 req?
TC-10	R8	CLOSING	OPEN	Safe transition; never both relays ON	State OPENING; OPEN relay = ON; CLOSE relay = OFF	Passed	R8 req R4 req?

TC-11	R8	OPENING	CLOSE	Safe transition; never both relays ON	State CLOSING; OPEN relay = OFF; CLOSE relay = ON	Passed	R8 req R4req?
TC-12	R9	Any moving state	Observe relay combination	Never OPEN+ CLOSE relay simultaneously	Never OPEN+ CLOSE relay simultaneously	Passed	R8 req?
TC-13	R10	OPENING	Wait >10 s without open limit	ERROR; relays OFF; error indication	No ERROR, state is OPENING more than 10s	Failed	Bug_2 R9 req?
TC-14	R10	CLOSING	Wait >10 s without close limit	ERROR; relays OFF; error indication	ERROR; relays OFF; error indication	Passed	R9 req?
TC-15	R11	Any valid state (STOPPED)	Activate both limits	ERROR; relays OFF; error indication	No Error	Failed	Bug_3 R10 req?
TC-16	R12	ERROR	OPEN	No movement; relays OFF	IGNORED: controller is in ERROR	Passed	R11 req?
TC-17	R12	ERROR	CLOSE	No movement; relays OFF	IGNORED: controller is in ERROR No movement; relays OFF	Passed	R11 req?
TC-18	R13	ERROR	RESET	STOPPED; relays OFF	STOPPED; relays OFF	Passed	R11 req?
TC-19	R14	CLOSED	Send invalid command, e.g. X	No movement; diagnostic message	No movement; UNKNOWN COMMAND	Passed	R13 req?
TC-20	R15	OPENING	Observe LEDs/relay	OPEN LED ON; CLOSE LED OFF; OPEN relay ON	OPEN LED ON; CLOSE LED OFF; OPEN relay ON; CLOSE relay OFF	Passed	R14 req?
TC-21	R15	CLOSING	Observe LEDs/relay	CLOSE LED ON; OPEN LED OFF; CLOSE relay ON	CLOSE LED ON; OPEN LED OFF; CLOSE relay ON; OPEN relay OFF	Passed	R14 req?
TC-22	R15	ERROR	Observe outputs	ERROR LED ON; relays OFF	ERROR LED ON; relays OFF	Passed	R14 req?
TC-23	R16	CLOSED	Single OPEN press	One OPEN command /one state transition	One OPEN command /one state transition	Passed	R16 req
TC-24	R16	CLOSED	Hold OPEN button	No uncontrolled repeated commands	No uncontrolled repeated commands	Passed	R16 req
TC-25 (Alexandra)	R16	CLOSED	Multiple OPEN press	One command is executed	After first press CLOSED -> OPENING Then IGNORED: OPEN is not valid in current state	Passed	R16 req
TC-26 (Alexandra)	R3	STOPPED	CLOSE	CLOSING	State CLOSING; CLOSE relay ON; OPEN relay OFF	Passed	R3 req
TC-27 (Alexandra)	R12	STOPPED/OPEN/OPENING/CLOSE/CLOSING	RESET	System state is not changed	System state is not changed	Passed	R12 req

Bug Report

https://docs.google.com/spreadsheets/d/1eu33SnrWu_8rkRPKTQjuD--UOsNzOOhA4d4Zigke0Uc/edit?usp=sharing

Session Summary

Total tests 27	Passed 23
Failed 3	Blocked 1
Bugs reported 3	Critical/High 3
Coverage notes	Build result Rejected for release